

Duration: 2.5 hours

MAX. MARKS 50

Section A [10 x 2 marks = 20 marks]
Attempt all questions in about 40-50 words.

MNIT JAIPUR

22CET112: ENGG DRAWING & SKETCHING

END TERM EXAMINATION 2024-25

SEM -II, SECTION - A, B & Backlog

TIME: 150 MIN

MAX MARKS: 50

PART A

Instructions: Attempt all questions in Part A on AutoCAD. Hand-sketching these questions will not fetch you any marks. All drawings should be printed on a single A4 sheet. Your name, ID, and Batch should be printed on the lower right-hand corner of the sheet. Upload the PDF file to server at 172.16.23.14. Make sure to sign the printout of your answer sheet before leaving the exam hall.

1. Prepare a title block on the bottom right-hand corner of sheet containing your name, ID, Section, Date, Course Name and Code. [1]
2. A line AB, 70 mm long is inclined at 30° to the H.P. Its end A is 12 mm above the HP and 20 mm in front of the VP. Its front view measures 50 mm. Draw the top view of AB and determine its inclination with the VP. [4]
3. A line AB, inclined at 40° to the VP, has its ends 50 mm and 20 mm above the HP. The length of its front view is 60 mm and its VT is 10 mm above the HP. Determine the true length of AB, its inclination with the HP and its HT. [5]
4. A circle of 55 mm diameter, rests on HP on a point A on the circumference, with its plane inclined 40° to the HP. Draw its projections if the top view of the diameter makes an angle of 30° with the VP. [10]
5. A pentagonal solid prism, having base side 35 mm and axis length 65 mm, has its axis inclined 40° to HP. One of the base sides lies in HP and is inclined 45° to the VP. Draw its projections. [15]

PART B

Instructions: Hand-sketch both the questions in this part on the given drawing sheets. Attempting these questions on AutoCAD will not fetch you any marks. All dimensions are in mm. You must mention your NAME, ID, and BATCH on top of the sheet on both sides.

6. Draw the top, front, and LHS view of the isometric sketch shown in below Fig. 1. [7]

P.T.O.

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MALVIYA NATIONAL INSTITUTE OF TECHNOLOGY JAIPUR
END-TERM EXAMINATION (50%), I & J (SLOT - 2), 2024 - 25
ENGINEERING DRAWING & SKETCHING (22CET - 112) on 3rd Dec 2024, 5:00 - 7:30 pm

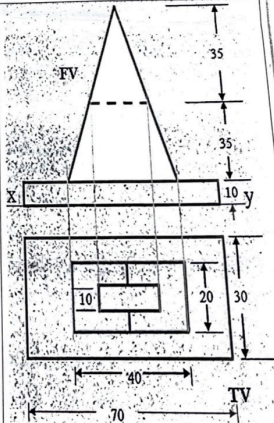
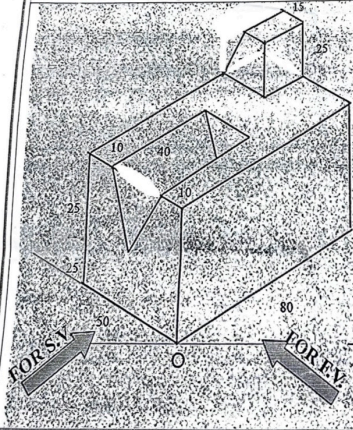
NOTE: 1. Answer ALL questions. MARKS are indicated in the bracket. ALL the dimensions are in millimetres.
 2. If any DATA is missing, assume the appropriate value & write your NAME, ID, Batch & SCALE.
 3. Default: 1- PROJECTION & Projection means draw ALL salient features. Upload the PDF file to <https://172.16.23.14>

Part A: AutoCAD (5:00 - 7:00 PM, 2 hours)

- Projectors drawn from HT and VT of line AB are 80mm apart, and those drawn from its ends are 50mm apart. End A is 10mm above H.P., VT is 55mm above H.P. while its HT is 35mm in front of V.P. Draw projections, locate traces, and find TL of line & inclinations with H.P. and V.P. [6 M]
- A $30^\circ - 60^\circ$ set square of the longest side 100mm long is in V.P. Its surface 45° inclined to V.P. One end of the longest side is 10mm, and the other is 35mm above H.P. Draw its projections. [9 M]
- A square pyramid with a 30mm base side & 50mm long axis rests on its apex on H.P., such that its one slant edge is vertical & a triangular face is perpendicular to V.P. Draw its projections. [8 M]
- A hexagonal pyramid base 25mm side and axis 55mm long has one of its slant edges on the ground. A plane containing that corner and the axis is perpendicular to the H.P. and inclined at 45° to the V.P. Draw its projections when the apex is nearer to the V.P. than the base. [12 M]

Part B: Sketching (7:00 - 7:30 PM, ½ hour)

- Draw the orthographic view of F.V., S.V., & T.V. for the isometric figure below. [7 M]
- Draw the isometric view from the orthographic figure as given below. [8 M]



ROUGH WORK (P.T.O)